OFFICIAL COLLEGE PROJECT ABSTRACT · PROJECT 31 OF 55

FLOOD RISK PREDICTION AND REAL-TIME EARLY WARNING SYSTEM USING MACHINE LEARNING AND GEOSPATIAL DATA

Domain: Geospatial Modeling & Disaster Intelligence | **Module:** project_31_flood_risk

OFFICIAL ABSTRACT

Flood Risk Prediction and Real-Time Early Warning System Using Machine Learning and Geospatial Data is a comprehensive and crucial project aimed at addressing unexpected natural disasters, public safety risks, and hydrological forecasting inefficiencies in modern urban and rural environments. As climate change and urbanization accelerate, communities increasingly rely on interconnected monitoring networks that continuously generate complex environmental metrics, making them prime targets for sudden catastrophic flood events if water level and precipitation surges are not proactively identified. Traditional monitoring methods often rely on scheduled or reactive reporting, which can be time-consuming, resource-intensive, and fundamentally inefficient against rapidly developing, complex hydrological anomalies and subtle topographic shifts. To address these persistent challenges and limitations, this study proposes an advanced, multi-layered approach combining cutting-edge deep learning techniques with intelligent machine learning algorithms for efficient, automated, and real-time environmental monitoring and flood risk prediction. The system utilizes specialized deep learning architectures to continuously capture and extract key behavioral features from extensive geospatial and temporal sensor streams, such as fluctuating river water levels, precipitation rates, soil moisture indexes, and topographical elevation deviations. Once these underlying structural patterns are mapped, the extracted features are fed into classification and forecasting algorithms designed to enhance decision-making, optimize computational overhead, improve prediction accuracy, and drastically reduce false alarm and misclassification rates. By profiling what constitutes normal, routine hydrological flow for each distinct geographic zone, the model establishes a dynamic baseline that adapts to changing climatic and seasonal load requirements over time. The proposed model is rigorously trained, validated, and tested on robust datasets encompassing diverse geospatial logs, operational river cycles, and simulated flood

inundation scenarios. This enables the framework to accurately distinguish between safe water flow and critical atmospheric and hydrological anomalies—such as flash floods, river overflows, and severe catchment saturation—with high precision and minimal latency. Experimental results and comprehensive performance evaluations demonstrate that the seamless integration of AI-driven feature extraction and machine learning models provides significantly improved classification and regression performance, resilience, and adaptability compared to traditional legacy forecasting approaches.

📌 Project Summary & System Specifications

Project Title	FLOOD RISK PREDICTION AND REAL-TIME EARLY WARNING SYSTEM USING MACHINE LEARNING AND GEOSPATIAL DATA
Specialization / Category	Geospatial Modeling & Disaster Intelligence
Target Technology Stack	Python (Flask), PyTorch / TensorFlow / Scikit-Learn, Modern Web UI
Default Backend Port	`:5031`
Local Implementation	`project_31_flood_risk`